Enriching a Knowledge Graph

Tempio Malatestiano in ArCo

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View on GitHub

Tempio Malatestiano

One of the key Renaissance works in Rimini and one of the most significant for the Italian Quattrocento

Project 1 · ArCo KG enrichment

Website: https://katyamugai.github.io/Project-Tempio-Malatestiano/index.html

Project Overview

What we investigated and why it matters

Selected resource

Tempio Malatestiano
Rimini, Italy

Main research problem

Relevant information is present in related photographic labels, but not as explicit RDF relations of the main resource.

Two gaps

- Internal architectural components
- Historical & artistic entities

ArCo IRI `https://w3id.org/arco/resource/ArchitecturalOrLandscapeHeritage/0800163046`

Info from photographic labels → **SPARQL evidence** → **explicit RDF triples**

Main idea: transform implicit knowledge into structured, machine-readable relations.

Methodology

A compact workflow from ArCo exploration to RDF enrichment

Key comparison: **direct RDF triples** ↔ **photographic labels**

1 → 2 → 3 → 4 → 5 → 6 → 7

1. Select ArCo resource
2. Explore RDF with SPARQL
3. Compare direct RDF vs labels
4. Identify 2 gaps
5. Test LLMs
6. Propose RDF triples
7. Create a website

Query 1 — Finding the main ArCo resource

```
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT DISTINCT ?cp ?label
WHERE {
  ?cp rdfs:label ?label .
  FILTER(REGEX(STR(?label), "Tempio Malatestiano", "i"))
}
LIMIT 50
```

Query 2 — Exploring the direct RDF description

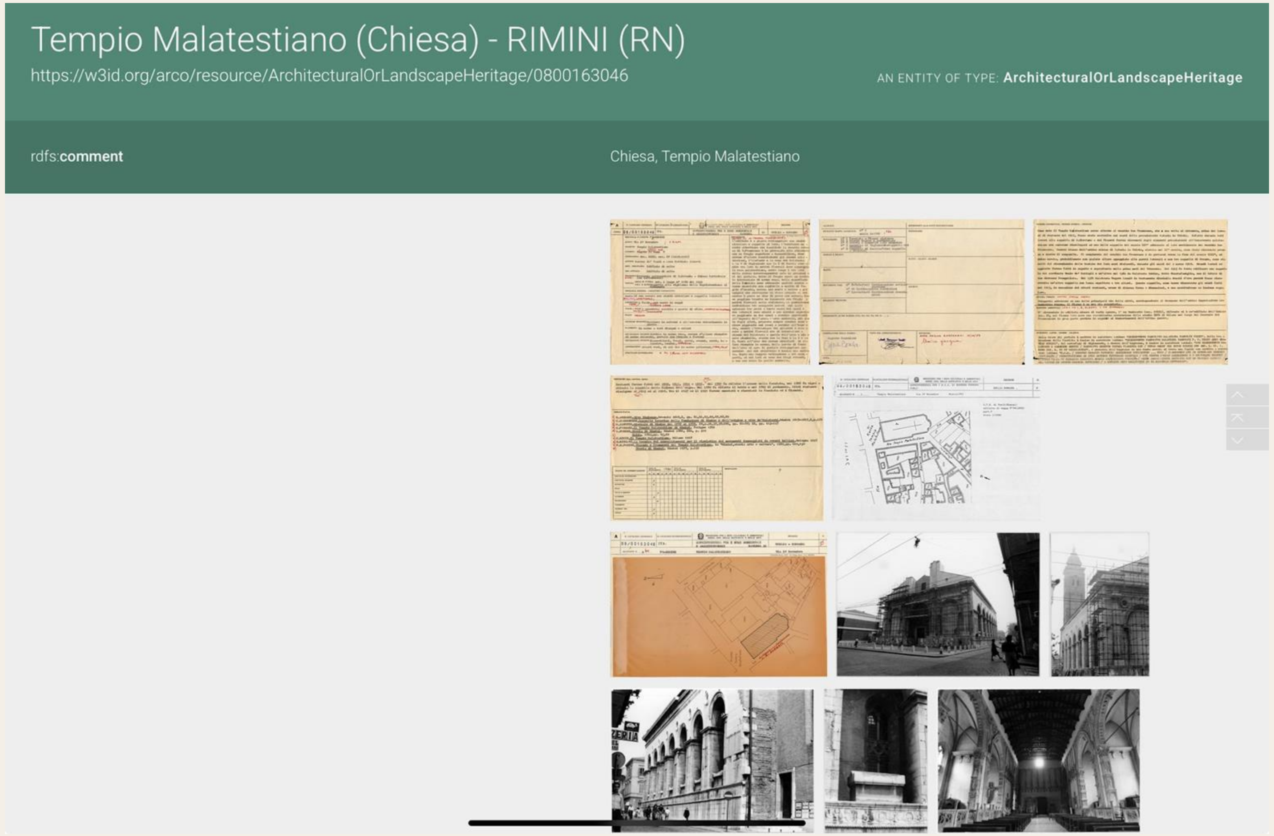
```
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT DISTINCT ?p ?pLabel ?o ?oLabel
WHERE {
  VALUES ?cp { <https://w3id.org/arco/resource/Architectura

  ?cp ?p ?o .

  OPTIONAL { ?p rdfs:label ?pLabel . }
  OPTIONAL { ?o rdfs:label ?oLabel . }
}
ORDER BY ?p
LIMIT 100
```

Gap 1 — Internal Architectural Components

The chapels exist in labels, but not as direct construction elements

Direct RDF result

Tempio → cdesc:hasConstructionElement → Facade



Query 3 — Checking existing construction elements

SPARQL Query

```
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX cdesc: <https://w3id.org/arco/ontology/construction-description/>

SELECT DISTINCT ?element ?elementLabel
WHERE {
  VALUES ?cp { <https://w3id.org/arco/resource/ArchitecturalOrLandscapeHe

  ?cp cdesc:hasConstructionElement ?element .

  OPTIONAL { ?element rdfs:label ?elementLabel . }
}
ORDER BY ?elementLabel
LIMIT 100
```

Results

element	elementLabel
https://w3id.org/arco/resource/Facade/0800163046-1	"Facade 1 of cultural property: 0800163046"@en
https://w3id.org/arco/resource/Facade/0800163046-1	"Facciata 1 del bene culturale: 0800163046"@it

The query returned only **one construction element**:

`https://w3id.org/arco/resource/Facade/0800163046-1`

The labels of this resource are:

- `Facade 1 of cultural property: 0800163046`
- `Facciata 1 del bene culturale: 0800163046`

Query 4 — Finding internal chapels in photographic resources

The chapels exist in labels, but not as direct construction elements

source	elementName	numberOfResources
Direct construction element	Facade	1
Photographic resource label	Cappella delle Virtù / S. Sigismondo	54
Photographic resource label	Cappella dello Zodiaco	43
Photographic resource label	Cappella degli Angeli	28
Photographic resource label	Cappella degli Antenati	26
Photographic resource label	Cappella d'Isotta	1

Photographic labels repeatedly mention

- Cappella delle Virtù / S. Sigismondo
- Cappella dello Zodiaco
- Cappella degli Angeli
- Cappella degli Antenati

Proposed enrichment

```
Tempio Malatestiano
  cdesc:hasConstructionElement
    ex:CappellaDelleVirtu ;
    ex:CappellaDelloZodiaco ;
    ex:CappellaDegliAngeli ;
    ex:CappellaDegliAntenati .
```

```
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX cdesc: <https://w3id.org/arco/ontology/construction-description/>

SELECT ?source ?elementName (COUNT(DISTINCT ?item) AS ?numberOfResources)
WHERE {
  VALUES ?cp { <https://w3id.org/arco/resource/ArchitecturalOrLandscapeHeritage/0800163046> }

  {
    ?cp cdesc:hasConstructionElement ?item .
    BIND("Direct construction element" AS ?source)
    BIND("Facade" AS ?elementName)
  }

  UNION

  {
    ?cp rdfs:seeAlso ?item .
    ?item rdfs:label ?photoLabel .

    BIND("Photographic resource label" AS ?source)

    BIND(
      IF(REGEX(STR(?photoLabel), "Cappella degli Antenati", "i"), "Cappella degli Antenati",
      IF(REGEX(STR(?photoLabel), "Cappella degli Angeli", "i"), "Cappella degli Angeli",
      IF(REGEX(STR(?photoLabel), "Cappella delle Virt", "i"), "Cappella delle Virtù / S. Sigismondo",
      IF(REGEX(STR(?photoLabel), "Cappella dello Zodiaco", "i"), "Cappella dello Zodiaco",
      IF(REGEX(STR(?photoLabel), "Cappella d.Isotta", "i"), "Cappella d'Isotta",
      ""))))) AS ?elementName
    )

    FILTER(?elementName != "")
  }
}
GROUP BY ?source ?elementName
ORDER BY ?source DESC(?numberOfResources)
```

Gap 2 — Historical & Artistic Entities

Entities are mentioned in labels, but not directly linked to the main resource

Entities found in photographic labels

Sigismondo Pandolfo Malatesta	historical figure
Isotta degli Atti	historical figure
Crocifisso giottesco	artistic object
Stemma Malatesta	heraldic element

Direct-link query result

No results

The entities are present as text, but not as structured RDF links.

Entity	Number of photographic resources
Sigismondo Pandolfo Malatesta	9
Sigismondo Malatesta	8
Isotta degli Atti	3
Statua di Sigismondo	2
Stemma Malatesta	2
Crocifisso giottesco	1

SPARQL Query 6

```
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT DISTINCT ?p ?pLabel ?o ?oLabel
WHERE {
  VALUES ?cp { <https://w3id.org/arco/resource/Architectura

  ?cp ?p ?o .

  OPTIONAL { ?p rdfs:label ?pLabel . }
  OPTIONAL { ?o rdfs:label ?oLabel . }

  FILTER(?p != rdfs:seeAlso)

  FILTER(
    REGEX(STR(?o), "Sigismondo", "i") ||
    REGEX(STR(?o), "Malatesta", "i") ||
    REGEX(STR(?o), "Isotta", "i") ||
    REGEX(STR(?o), "Giotto", "i") ||
    REGEX(STR(?o), "Crocifisso", "i") ||
    REGEX(STR(?o), "stemma", "i") ||
    REGEX(STR(?oLabel), "Sigismondo", "i") ||
    REGEX(STR(?oLabel), "Malatesta", "i") ||
    REGEX(STR(?oLabel), "Isotta", "i") ||
    REGEX(STR(?oLabel), "Giotto", "i") ||
    REGEX(STR(?oLabel), "Crocifisso", "i") ||
    REGEX(STR(?oLabel), "stemma", "i")
  )
}
ORDER BY ?p
LIMIT 100
```

Results

p	pLabel	o	oLabel

Final property mapping

core:involvesAgent | **a-dd:hasAssociatedObject** | **a-dd:hasElementAffixedToCulturalProperty**

RDF Enrichment — Reuse First

New triples, existing ontology properties

Key modelling decision

No new local properties were needed. The enrichment introduces local resources, but reuses ArCo/Core relations.

LLMs + Final Takeaway

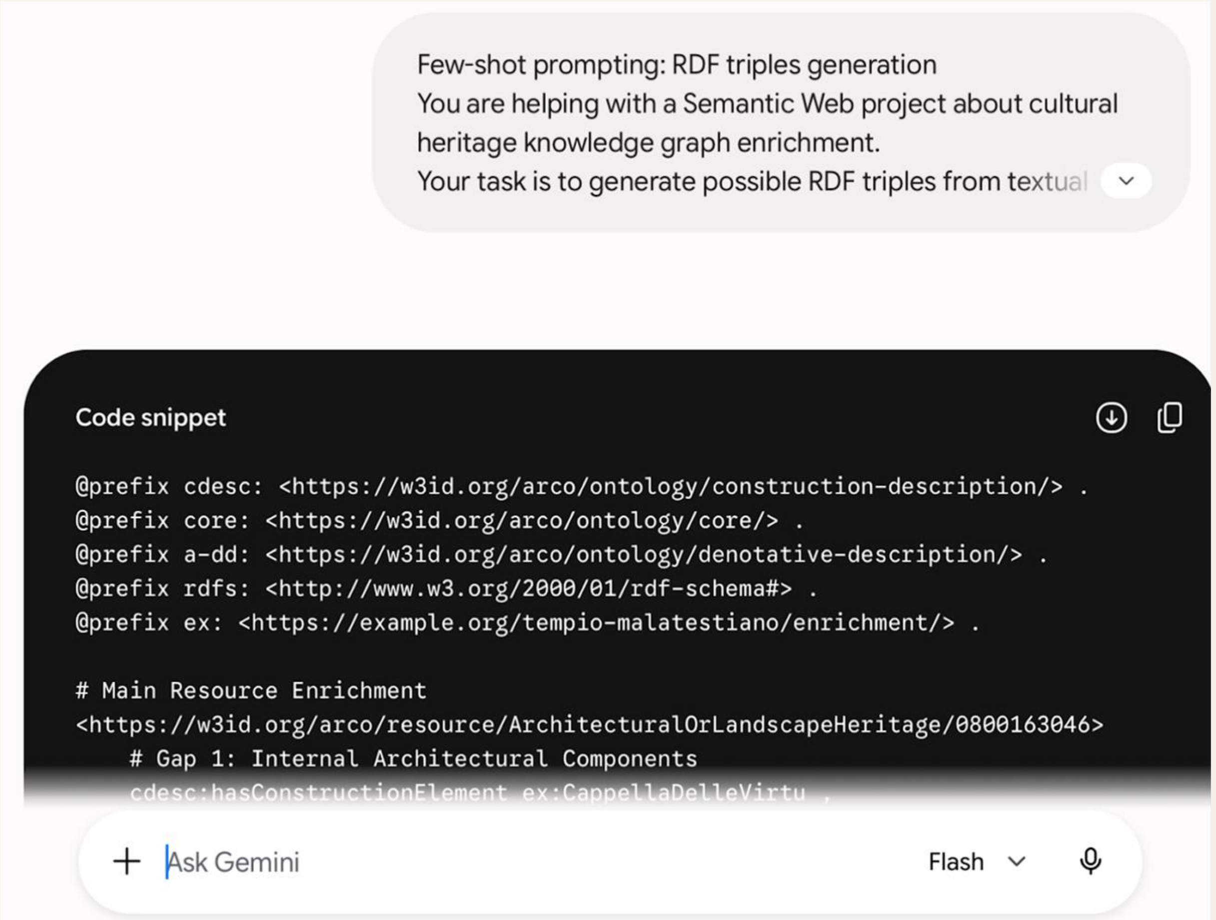
LLMs supported the workflow, but SPARQL evidence drove the final model

Prompting techniques

- Zero-shot: useful for orientation
- Chain-of-thought: better gap reasoning
- Few-shot: best for RDF generation

Model comparison

ChatGPT: stronger explanation
Gemini: cleaner Turtle output



Challenges + Final Takeaway

Challenges we faced and the conclusion

Challenges

- Understanding the difference between direct RDF data and indirect information
- Extracting useful information from photographic resource labels and creating proper SPARQL queries
- Choosing existing ontology properties instead of creating new ones
- Presenting SPARQL, RDF and everything clearly on the website

Final takeaway

The final enrichment proposal makes the representation of Tempio Malatestiano more explicit, structured and directly queryable. After the proposed enrichment, it would be possible to query the monument directly for its internal chapels, associated historical agents, related artistic object and affixed heraldic element. This is more effective than searching for these entities only inside photographic resource labels.